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Artificial Intelligence tool for prediction of ECM mimics hydrogel formulations via click chemistry☆

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ABSTRACT

A user-friendly machine learning (ML) predictive tool is reported for designing extracellular matrix (ECM)-mimetic hydrogels with tailored rheological properties. Developed for regenerative medicine and 3D bioprinting, the model leverages click chemistry crosslinking to fine-tune the mechanical behaviour of gelatin- and hyaluronic acid-based hydrogels. Using both experimental rheological data and synthetic datasets, our supervised ML approach accurately predicts hydrogel compositions, significantly reducing the cost and time associated with trial-and-error approach. Despite advancements in the field, existing models remain limited in their ability to mimic the ECM due to the use of non-natural polymers, reliance on a single type of biologically active macromolecule, and physical crosslinking reactions with limited tuneability. Additionally, their lack of generalizability confines them to specific formulations and demands extensive experimental data for training. This predictive platform represents a major advancement in biomaterial design, improving reproducibility, scalability, and efficiency. By integrating rational design, it accelerates tissue engineering research and expands access to customized ECM-mimetic hydrogels with tailored viscoelastic properties for biomedical applications, enabling both experts and non-experts in materials design.

1. Introduction

Hydrogels are highly versatile materials with a wide range of applications, including tissue engineering, drug delivery, and biosensors, due to their unique properties such as high-water content, tunable mechanical properties and biocompatibility [51,52]. In particular, their ability to closely mimic the extracellular matrix (ECM) makes them ideal materials for biomedical applications that require an environment supporting cell adhesion, proliferation, and differentiation [1]. However, the design and optimization of hydrogel formulations, with specific desirable biochemical and physical properties (e.g. elasticity, viscoelasticity, strength) remains challenging. Indeed, experimental parameters involved in hydrogel fabrication are numerous, such as polymer types and ratio, functionalization degree, crosslinking percentage and reaction conditions, thus making it difficult to correlate quantitatively the contribution of all variables on the final material properties [2].

Therefore, a process optimization would be desirable particularly in hydrogel formulations used in 3D bioprinting, where a proper balance between ink printability and biocompatibility must be achieved. As a matter of fact, printability largely depends on the hydrogel's rheological properties, especially on their shear-thinning behaviour and viscoelasticity. The combination of both will affect how smoothly the hydrogel flows through the printer nozzle without clogging, its ability to maintain structural integrity post-deposition, as well as shear stress values that can be detrimental to cell viability (i.e. percentage of percentage of surviving cells after printing) [3–5,53].

In addition, hydrogel's rheological properties are also fundamental for its final applications in tissue engineering [6]. Indeed, it is now common knowledge that cell behaviour is very much influenced by the mechanical properties of their microenvironment. A microenvironment with an optimal balance between elasticity and viscosity more efficiently promotes cell differentiation and functional tissue formation,

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enhancing tissue regeneration and increasing possibilities for clinical translation [7–9]. Therefore, in order to closely recapitulate the properties of human tissues, it is of paramount importance to have complete control over the elastic response (G′ or storage modulus) and viscous response (G″ or loss modulus) of a material employed as a scaffold.

Traditionally, researchers have relied on a trial-and-error approach to identify optimal ink formulations, which often involves numerous experiments to achieve the desired physical and biochemical properties. This process is not only costly and time-expensive but also frequently inefficient. The ability to perform reverse design engineering (i.e. formulation of hydrogels with predetermined mechanical properties), is a critical aspect for the development of ECM mimetics inks.

To address these limitations, the advent of artificial intelligence (AI), and specifically machine learning (ML), paved the way for a new paradigm in research approach for several biomedical fields [54,55]. As a matter of fact, ML can be used to create predictive models able to map complex relationships between multiple input parameters and output properties, enabling the prediction of hydrogel properties based on a pilot data set [13,14,18,20].

To the best of our knowledge, examples present in the literature in which AI has been employed for hydrogel formulation prediction are not so many, and have been applied to materials which are made of synthetic polymers [12,22], a single biopolymer type [16,17,19] and crosslinked via photopolymerization or physical crosslinks [15,17,21,23,56]. Despite these advances, the existing models in literature are highly constrained in their capacity to mimic the ECM because of the employment of non-natural polymers, just one type of macromolecule biologically active, and the use of non chemoselective reactions. Furthermore, most models lack generalizability, since they are often tailored to specific formulations with narrow application domains, and require large amount of experimental pilot data to build up the model.

To overcome the current limitations, in this work, we present the development of a user-friendly ML-based tool aimed at predicting the formulations of multifunctional hydrogels, using the desired rheological parameters as input (i.e. G′ and G″) (Fig. 1). To replicate essential properties of the ECM, we selected hyaluronic acid (HA) and gelatin (a hydrolysed derivative of collagen) as bioactive components and polyethylene glycol (PEG) star as crosslinker agent. Two well-known click chemistry reactions, namely (SPAAC) and (DA), were used to covalently link together all polymers and thus form the 3D network of the hydrogels. The ML model was built from a pilot dataset where materials quantities (mg) were associated to their viscoelastic properties (G′ and G″). Importantly, to limit experimental data need, a synthetic dataset was generated and specific constraints were incorporated (i.e. limited ranges materials quantities). Results show that the ML model was capable of accurately predicting random hydrogel formulations when given the storage and loss modulus as input, for both click reactions. The predictive tool that we introduce represents a significant advancement in biomaterial science because it enhances material design efficiency by significantly reducing experimental time and material waste. Moreover, the model's accurate predictions with limited datasets addresses a critical need in biomaterial research, where obtaining extensive data is often challenging. We do believe that our AI-driven tool holds the potential to drastically speed up research in tissue engineering and opens the possibility to non-experts in material design to generate an ECM-mimetic hydrogel with custom viscoelastic properties.

2. Materials and methods

2.1. Materials

Gelatin from porcine skin type A (G2500, Sigma Aldrich), N-hydroxysuccinimide (NHS), phosphate buffered saline (PBS), N_3-PEG_4-COOH, 4arm-PEG20K-DBCO (4arm-PEG-DBCO), $NaBH_3CN$, N_3-$PEGNH_2$, 5-methylfurfural, 5-methylfurfurylamine, 4arm-PEG10K-Maleimide (4arm-PEG-MA) were purchased from Sigma-Aldrich. 1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide (EDC), 4-(4,6-Dimethoxy-1,3,5-triazin-2-yl) (DMTMM) were purchased from TCI; Hyaluronic acid (YH05852 average MW 1.0–2.0 million Da) from Carbosynth Ltd.

2.2. Azide functionalized gelatin (gel-N_3)

Gelatin was functionalized with an azide group as reported in [26] obtaining Gel—N_3. Briefly, 250 mg of Gelatin type A (0.085 mmol) was dissolved in 10 mL of PBS 5.5 in a falcon. Meanwhile, N_3-PEG_4-COOH (50 mg, 0.171 mmol) was dissolved in PBS 5.5 (5 mL) in a bottom round flask at rt. Then, EDC (66 mg, 0.342 mmol) and NHS (40 mg, 0.342 mmol) were added under gentle stirring. The gelatin solution was added to the PEG solution after 30 min. The reaction was stirred overnight at

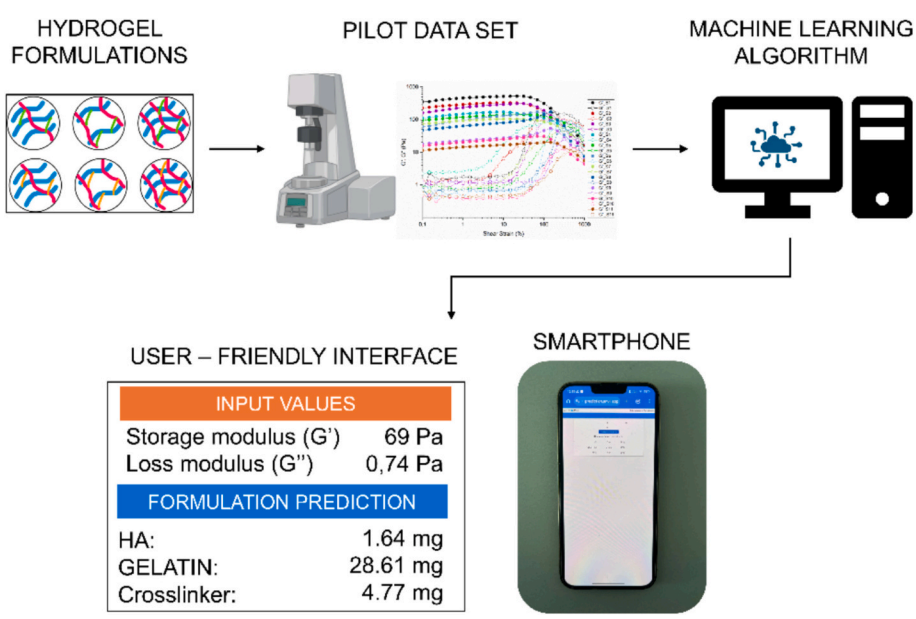

Fig. 1. Workflow of the development of machine learning algorithm to predict hydrogel formulations with specific rheological properties able to mimic ECM, using a user-friendly graphical interface.

37 °C. Further EDC (66 mg, 0.342 mmol) and NHS (40 mg, 0.342 mmol) were added, and the mixture was stirring for 4 h, then dialyzed with a 14 kDa cellulose membrane against NaCl 0.1 M for 1 day and then against milli-Q water for 2 days. The obtained azidated gelatin ($GelN_3$) was lyophilized and stored at −20 °C before usage.

2.3. Azide functionalized hyaluronic acid (HA-N_3)

Hyaluronic acid was functionalized with an azide group as reported in [26] HA-N_3. Briefly, 100 mg (0.264 mmol) of HA were dissolved in 20 mL MES buffer pH 5.5 for 3 h. DMTMM (146 mg, 0.572 mmol) was added. Then, N_3-PEG-NH_2 (115 mg, 0.572 mmol) was added after 30 min and the solution was stirred overnight, dialyzed with a 14 kDa cellulose membrane against NaCl 0.1 M for 2 days and then against milli-Q water for 2 days. The obtained HA-N_3 was lyophilized and stored at −20 °C before usage.

2.4. Furan functionalized gelatin (gel-5MF)

Gelatin was factionalized with 5-methylfurfural as similarly reported in [27]. Gelatin type A (1.00 g) was dissolved in 30 mL of citric acid buffer pH 6 and heated at 37 °C until a homogeneous solution was obtained. Then 3,4 mL of 5-methylfurfural was added and left under gentle stirring at 37 °C. After 30 min, 1,07 g of $NaBH_3CN$ was added and the solution was stirred stirring for 3 h. The solution was dialyzed against a NaCl solution (0.1 M) for 1 day, followed by distilled water for 4 days, using 14 kD dialysis membranes at 37 °C. The obtained solution was freeze-dried and keep at −20 °C.

2.5. Furan functionalized hyaluronic acid (HA-5MF)

1 g of hyaluronic acid was dissolved in 100 mL of MES buffer pH 5.5 overnight under stirring. Then 276 mg of DMTMM were added. After 30 min 54 μL of 5-methyl furfurilammine were added drop by drop and the solution were stirred overnight. The solution was dialyzed against a NaCl solution (0.1 M) for 1 day, followed by distilled water for 4 days, using 14 kD dialysis membranes at 37 °C. The obtained solution was freeze-dried and keep at −20 °C.

2.6. NMR analysis

^{1}H NMR were performed using a Varian Mercury 400 M Hz. Sample were dissolved in D_2O with 0.05 % (wt%) TMSP at a concentration of 8 mg/mL.

2.7. SPAAC hydrogels preparation

Different amounts of HA-N_3, Gel—N_3, and 4armPEGDBCO were separately dissolved in PBS (pH 7.4) to reach the final concentrations shown in Table 1, achieving a total hydrogel volume of 2 mL. Each solution was maintained at 40 °C with continuous stirring at 1200 rpm for one hour to ensure complete dissolution. Afterward, the gelatin solution was combined with the hyaluronic acid solution. The required amount of linker was then added to this mixture, forming the hydrogel under stirring at 800 rpm at 40 °C in few minutes.

Table 1
SPAAC hydrogels formulations.

HA-N_3 (mg/mL)	Gel-N_3 (mg/mL)	4arm-PEG-DBCO (mg/mL)
3.7	26.3	10.0
2.2	28.1	8.4
2.0	28.0	10.0
2.0	28.0	5.0
1.5	27.3	8.7
1.3	20.7	7.9
1.3	17.7	5.5
1.3	28.7	10.0
1.3	18.7	10.0
1.3	18.7	6.7
1.3	18.7	3.3

2.8. DA hydrogels preparation

Different amounts of HA-5MF, Gel-5MF, and 4armPEGMA were separately dissolved in PBS (pH 7.4) to reach the final concentrations shown in Table 2, achieving a total hydrogel volume of 2 mL. Each solution was maintained at 40 °C with continuous stirring at 1200 rpm for one hour to ensure complete dissolution. Afterward, the gelatin solution was combined with the hyaluronic acid solution. The required amount of linker was then added to this mixture, forming the hydrogel under stirring at 800 rpm at 40 °C in about 3 h.

2.9. Rheological analysis

Rheological characterization was performed with a Modular Compact Rheometer MCR 92 (Anton Paar, GmbH, Graz, Austria) equipped with a plane plate (diameter = 50 mm). About 2 mL of each hydrogel were dissolved in PBS (pH = 7.4) at thermomixer (rpm = 1200, T = 40 °C). Amplitude sweeps tests (shear strain (γ) range from 0,1 to 1000 %; frequency ($\dot{\gamma}$) = 10 Hz) were performed by setting a height gap of 1 mm and a temperature of 40 °C. This last parameter was specifically chosen to resemble the printing temperature. Data were elaborated with RheoCompass™ software.

2.10. 3D printing

An STL file was prepared with the CAD software Fusion360 (Autodesk) and the slicing was performed with slic3r.

Extrudability and 3D printing accuracy of predicted hydrogel formulations were tested with a BIOX 3D printer (CELLINK®). All formulations were printed in 70 mm petri dish with a pneumatic preheated at 40 °C. A 3 mL syringe was equipped with a 22 G conical nozzle (CELLINK®). The printing speed was set to be 2,0 mm/s.

Pictures of the 3D constructs were taken with a digital camera right after printing and strand thickness (d) measurements were performed with the ImageJ software and an ink spreading factor (Sp) was calculated as follows:

$$Sp\% = \frac{Tr - Tn}{Tn} \times 100$$

where Tr is the real strand thickness measured and Tn is the nominal thickness of the strand (1 mm).

A drop test was performed for each formulation in order to determine the optimal printing pressure (i.e. the pressure at which the ink started to flow constantly throughout the nozzle).

2.11. AI machine learning model

A neural network model (NNM) was developed to evaluate the

Table 2
DA hydrogels formulations.

HA-5MF (mg/mL)	Gel-5MF (mg/mL)	4arm-PEG-MA (mg/mL)
3.7	26.3	10.0
3.0	27.0	15.0
3.0	27.0	10.0
2.7	28.0	10.8
2.0	28.0	15.0
2.0	28.0	10.0
2.0	28.0	5.0
1.3	28.7	10.0
1.3	18.7	10.0

printability of hydrogel formulations based on user-defined input parameters. The model receives the formulation parameters as input and predicts the rheological properties. Since no predefined mathematical framework exists for determining the optimal neural network architecture (i.e., number of layers, neurons per layer, epochs, and activation function), the model design was refined through an iterative hyperparameter tuning process.

To optimize model performance, various neural network configurations were trained and evaluated using k-fold cross-validation. In this approach, the dataset was divided into *k* subsets (folds). In each iteration, one subset was reserved for testing, while the remaining *k*-1 subsets were used for training. This process was repeated *k* times, ensuring that each fold served as a test set once. This method mitigates overfitting and provides a more reliable estimate of the model's generalization performance. Given the limited size of the available dataset (Tables S1 and S2, Supporting information), all input features were normalized to enhance numerical stability and improve model convergence. Data normalization was performed to scale features within a defined range, preventing any single feature from disproportionately influencing the learning process. To account for the dataset limitations, the range of hyperparameter values was carefully selected to prevent overfitting while still allowing the model to explore different configurations. A randomized search approach efficiently sampled from these predefined hyperparameter ranges, balancing computational feasibility with model optimization. Batch sizes, learning rates, and hidden layers counts were selected based on preliminary experiments to ensure stable training and meaningful generalization. The model hyperparameters were selected using random search within predefined ranges (Table 3). This systematic exploration of hyperparameters enabled the identification of an optimized network architecture for hydrogel G′ and G″ prediction. The classification model was trained using the binary cross-entropy loss function, which is well-suited for binary classification tasks and ensures that the model learns a probability-based decision boundary for predicting rheological profile.

A second neural network model was trained to predict hydrogel formulations based on their viscoelastic properties, specifically storage modulus (G′) and loss modulus (G″). The model structure was optimized using the same random search approach, exploring various hyperparameter combinations to identify the best-performing configuration (Table 3). As with the classification model, all input features were normalized before training to ensure consistency and improve numerical stability. The model was trained using the Mean Squared Error (MSE) loss function, which is widely used for regression tasks to minimize the average squared difference between predicted and actual values.

By leveraging this inverse modeling approach, the trained neural network estimates potential hydrogel formulations with desired viscoelastic properties. This enables a data-driven formulation design process, reducing the need for extensive experimental trial and error.

3. Results and discussion

In this study, we present the development of a machine learning-based tool designed to predict hydrogel formulations from their viscoelastic properties, specifically storage modulus (G′) and viscous (G″) modulus.

Gelatin and hyaluronic acid were chosen as the hydrogel's biopolymeric components. The rationale behind such choice relies on the fact that gelatin is derived from collagen, the most abundant protein within the body, while HA is a ubiquitous polysaccharide of the ECM [1,2,28]. Moreover, there are countless examples in the literature of hydrogels made of gelatin and/or HA successfully employed in a broad spectrum of biomedical applications, spanning from regenerative medicine [29–31], disease modeling [2,32,33], biomedical devices [34,35], drug testing [36–38], and many more [39–41]. Gelatin and HA (Fig. S1) were covalently crosslinked using tetrafunctional star PEG via two click chemistry reactions: strain-promoted azide-alkyne cycloaddition (SPAAC) and Diels-Alder cycloaddition (DA) (Fig. 2). The use of click chemistry, providing precise covalent crosslinking control, further enhances the complexity and potential of this approach, enabling fine-tuned hydrogel properties crucial for tissue mimicry. In particular, SPAAC and DA reactions are extensively employed in hydrogel synthesis due to their efficiency, biocompatibility, and ability to form stable network structures without the need for potentially toxic catalysts, making them ideal for the creation of customized and optimized formulations for tissue engineering applications [2,42].

Different stoichiometries and experimental conditions were adopted, and the viscoelastic properties of the obtained hydrogel formulations were tested via rheological analysis to generate a pilot dataset.

3.1. Hydrogel viscoelastic properties characterization

Hydrogels exhibit complex viscoelastic behaviour, characterized by their storage and loss moduli. Amplitude sweep tests at a fixed frequency help identify the linear viscoelastic range (LVR), where the storage modulus (G′) remains stable under strain, distinguishing the material's elastic and viscous components.

To prepare the pilot dataset, hydrogels were formulated by varying the overall content of gelatin, hyaluronic acid, and crosslinker, as well as their reciprocal ratio. For each formulation, viscoelastic properties were characterized in terms of G′ and G″ values by performing amplitude sweep tests (Tables 4 and 5) (Fig. 3).

For all hydrogels tested, the storage modulus (G′) is approximately an order of magnitude higher than the loss modulus (G″), indicating that their behaviour is primarily governed by elasticity rather than viscosity within the LVR [6]. This viscoelasticity, characterized by both viscous flow and elastic shape retention, is influenced by the formulation specifics, as seen when comparing SPAAC and DA hydrogels with similar component ratios. In SPAAC hydrogels, G′ absolute values range from ~10 to 450 Pa, significantly higher than the ~1 to 25 Pa range in DA hydrogels. This difference can be attributed to the difference lengths of linkers (20 kDa in SPAAC and 10 kDa in DA) and a higher functionalization degree of hyaluronic acid in SPAAC hydrogels (22 %) compared to DA hydrogels (5 %) (Fig. S2, Supporting information). In fact, a more stretchable hydrogel network (conferred by longer linkers) and a higher crosslinking density (conferred by higher degree of functionalization), enhance SPAAC material's ability to store elastic energy and retain its shape during deformation. Conversely, a shorter linker and lower crosslinker density in DA hydrogels reduce crosslinking sites, yielding a less elastic network and a narrower range of G′ values even if materials ratios are changed.

On the other hand, neither polymer concentration nor crosslinker density and molecular weight, seem to influence the determination of

Table 3
Hyperparameters for SPAAC and DA models for the first and second neural network models.

Hyperparameter	First neural network model	Second neural network model
	Values	Values
Batch size	1, 11	1, 8, 16, 32
Epochs	10, 20, 30	10, 20, 30, 50, 100
Hidden layer size	2, 3, 5	2, 3, 4, 5
Learning rate	0.0001, 0.001, 0.005	0.0001, 0.001, 0.005, 0.01, 0.1

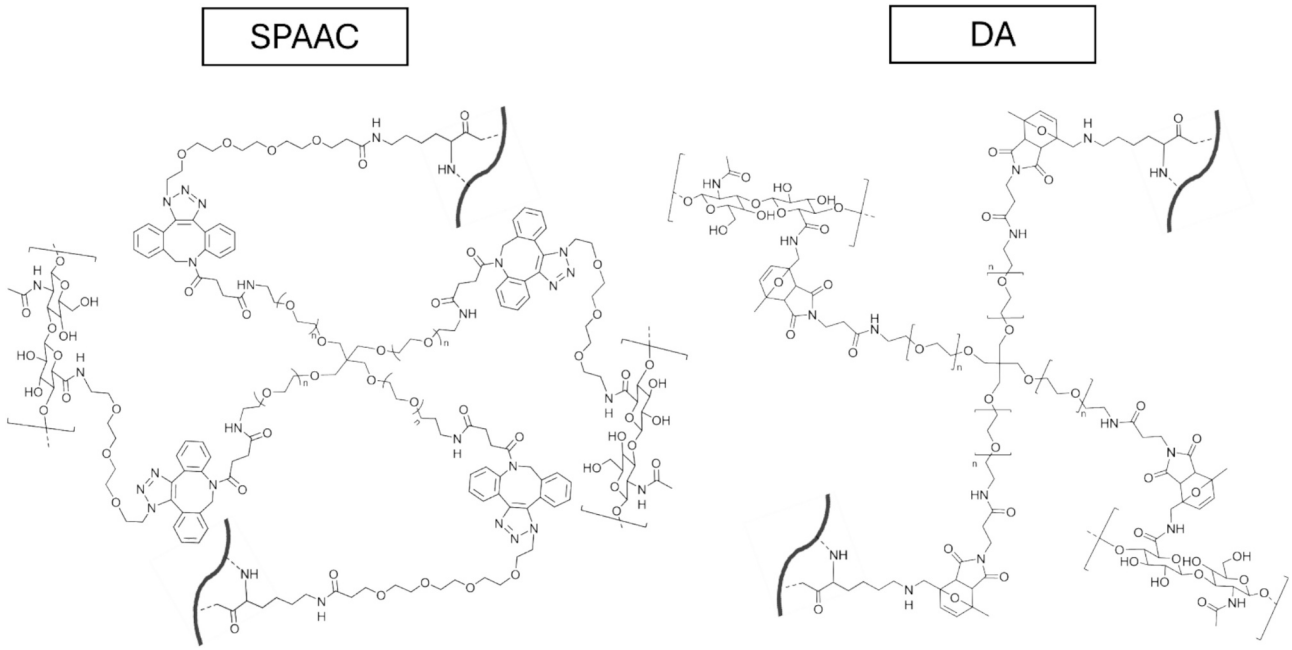

Fig. 2. Illustration of SPAAC and DA hydrogels network.

Table 4
G′ and G″ values of SPAAC hydrogel formulations obtained through amplitude sweeps tests.

	HA-N_3 (mg/mL)	Gel-N_3 (mg/mL)	4arm-PEG-DBCO (mg/mL)	G′ [Pa]	G″ [Pa]
S1	3.7	26.3	10.0	458.0	1.7
S2	2.0	28.0	10.0	286.5	1.3
S3	1.5	27.3	8.7	231.3	1.1
S4	2.2	28.1	8.4	146.3	2.5
S5	1.3	28.7	10.0	118.1	0.8
S6	1.3	20.7	7.9	96.4	0.7
S7	2.0	28.0	5.0	91.5	0.8
S8	1.3	18.7	10.0	64.0	0.7
S9	1.3	18.7	6.7	22.6	1.2
S10	1.3	17.7	5.5	20.3	0.4
S11	1.3	18.7	3.3	14.7	0.4

Table 5
G′ and G″ values of DA hydrogel formulations obtained through amplitude sweeps tests.

	HA-5MF (mg/mL)	Gel-5MF (mg/mL)	4arm-PEG-MA (mg/mL)	G′ [Pa]	G″ [Pa]
D1	3.0	27.0	15.0	25.1	1.9
D2	2.0	28.0	15.0	23.4	2.0
D3	3.7	26.3	10.0	22.4	3.2
D4	2.7	28.0	10.8	16.8	2.4
D5	3.0	27.0	10.0	14.7	2.1
D6	1.3	28.7	10.0	8.0	1.3
D7	2.0	28.0	10.0	6.9	1.5
D8	2.0	28.0	5.0	3.7	1.1
D9	1.3	18.7	10.0	3.0	0.8

the loss modulus, since the two data set present similar G″ values, ranging from ~0.5 to 2 Pa in SPAAC hydrogels and ~0.5 and 3 Pa in DA hydrogels.

From amplitude sweep analysis no crossover point was observed across the entire range of shear strain tested (γ from 0.1 to 1000) in DA hydrogels. Conversely, under the same shear conditions, SPAAC hydrogel formulations reached a crossover point at shear strain around 215 % (Fig. S2, Supporting information). This difference can likely be attributed to the influence of linker length on the onset of non-covalent interactions inside the material 3D structure. More specifically, DA hydrogels incorporate shorter PEG linkers, leading to a more compact structure. The proximity of functional groups on polymer chains favours the onset of non-covalent crosslinking (e.g. hydrogen bonds, electrostatic interactions) creating a reinforced network that better resists deformation, thus preventing the onset of a crossover point in the tested shear strain (Fig. S3, Supporting information). In contrast, SPAAC hydrogels have longer linker chains that leave the overall network more flexible and less tight. As shear strain increases, chains are allowed to stretch until the structure eventually breaks (crossover point) [43,44].

3.2. Machine learning model development and training

A first neural network model was trained to assess the printability of a given hydrogel formulation provided by the user. The model takes the user's input and provides an evaluation of the rheological profile of the hydrogel formulation.

In an initial effort to develop such a dataset, a neural network was implemented to predict rheological properties based on specific hydrogel formulations. This network, trained on a limited initial dataset, serves as a foundational tool for generating synthetic data that aligns with experimentally observed trends. Due to the limited availability of real data from laboratory experiments and the restricted accessibility of literature, generating a synthetic dataset became a reasonable approach. It allows for obtaining a more extensive and usable dataset for training the final model while attempting to avoid overfitting. To enhance the reliability and relevance of the synthetic dataset, specific constraints were incorporated, including limited ranges for the quantities of HA, GEL, and linker (Tables S1 and S2, Supporting information). These thresholds reflect the practical boundaries within which hydrogel formation is possible and/or useful; formulations with components below or above these limits cannot yield functional hydrogels.

The relevance of each data and its correlation was identify using a correlation matrix setting statistical significance at $p < 0.05$ (Fig. S4, Supporting information). From the correlation matrix in Fig. 4A, it is possible to extract some information regarding the correlation between the inputs and outputs of the SPAAC model: (i) A strong positive correlation between HA and G′. (ii) A moderate correlation for both gelatin

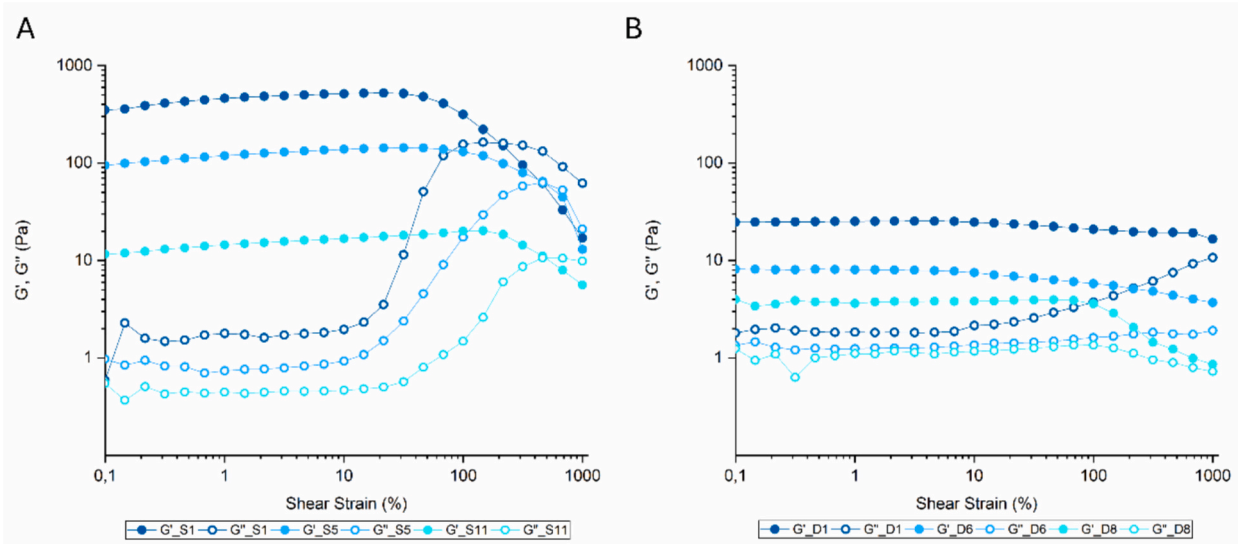

Fig. 3. Representative examples of A) SPAAC and B) DA hydrogels amplitude sweeps analysis.

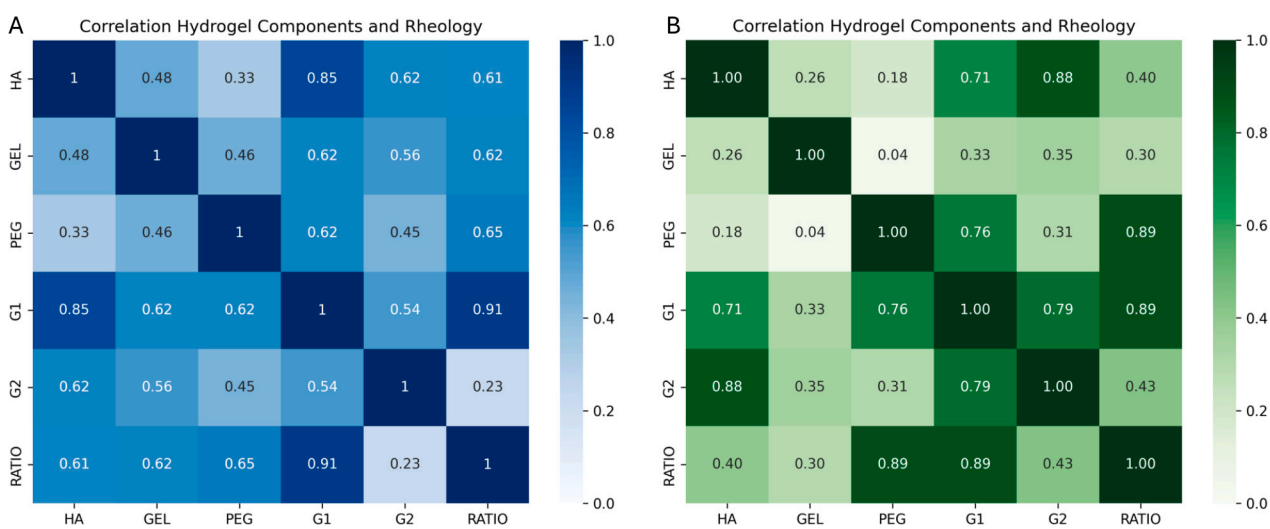

Fig. 4. Correlation matrix of SPAAC (A) and DA (B) ML models. G1 = storage modulus, G2 = loss modulus, PEG = linker, GEL = gelatin, HA = hyaluronic acid, RATIO = G′/G″.

and linker with the elastic modulus. (iii) A moderate correlation for each input of the model with G″. The DA correlation matrix (Fig. 4B) instead reveals the following key relationships among the variables: (i) A strong positive correlation between HA and G″. (ii) A moderate correlation between HA and G′. (iii) A strong positive correlation of the linker with G′.

In conclusion, these correlation matrices highlight the importance of HA and PEG as critical variables for predicting rheological properties of the proposed hydrogel formulations. In particular, for the SPAAC dataset, HA content proved to be the primary factor influencing G′ when comparing formulations with the same amount of material. On the other hand, for DA dataset, HA influenced the viscous properties more than the elastic ones. It was indeed the contribution of the linker that proved to have a greater impact in determining G′, suggesting that crosslinking is the key factor affecting material elasticity. In contrast, gelatin appears to have a minimal impact on both datasets.

3.3. Machine learning model validation

To assess the performance and accuracy of regression models, specific error metrics were used. The comparison in Table 6 highlights the performance differences between the SPAAC and DA predictive models based on key error metrics for hydrogel rheology evaluation. The DA model demonstrated lower error values across most metrics, indicating superior predictive accuracy. Specifically, the DA model achieved a lower Mean Squared Error (MSE) for G′ (14.34 Pa vs. 154.84 Pa) and G″ (0.27 Pa vs. 0.042 Pa) compared to the SPAAC model, suggesting improved precision in predicting storage and loss moduli. The Mean Absolute Percentage Error (MAPE) was lower for the DA model (25.10 % for G′ and 16.43 % for G″) compared to the SPAAC model (26.64 % for G′

Table 6
Error metrics for SPAAC and DA predictive models.

Metric	SPAAC		DA	
	G′ [Pa]	G″ [Pa]	G′ [Pa]	G″ [Pa]
MSE	154.84	0.042	14.34	0.27
RMSE	12.44	0.204	3.79	0.53
MAE	9.97	0.14	3.07	0.37
MAPE	26.64 %	18.15 %	25.10 %	16.43 %

and 18.15 % for G″), reflecting higher relative accuracy. Finally, the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) together suggest that errors for G′ may not be uniformly distributed, as RMSE is higher than MAE. This indicates the presence of outliers or large deviations for specific data points in both models. For G″, the closer alignment of RMSE and MAE implies more consistent errors.

The error metrics demonstrate a performance gap between G′ and G″, which might arise from differences in their underlying physical principles. G′ often reflects solid-like behaviour and is influenced by subtle changes in the material's microstructure, which may be harder to model. In contrast, G″ representing viscous behaviour might be governed by simpler rheological laws that the model captures better or more accurately in comparison to G′, where the relative error is smaller.

Overall, results suggest that improvements are needed to reduce the errors and to enhance the model's overall accuracy, especially for G′. This might include refining the model, identifying additional features, or incorporating more data. Despite all that, it is possible to say that the combined performance across outputs suggests the models are generally reliable, making them suitable for the prediction of hydrogel formulations from their rheological properties. Furthermore, compared to the state of the art, this approach can achieve accurate predictions even with a limited dataset, a significant advancement over typical ML models in materials science that often necessitate extensive data for training. The data efficiency is achieved through the human knowledge supported by AI's capacity, to learn fundamental chemical principles and reaction dynamics, enabling robust generalization to similar systems with minimal training data.

Validation of the machine learning models was performed by selecting values of G′ and G″ from which to compute the hydrogel formulations. Importantly, G′ and G″ values were chosen to span across the orders of magnitude present in the pilot dataset. Afterwards, the resulting hydrogel formulations have been tested with amplitude sweep tests to compare experimental data with the predicted ones (Tables 7 and 8).

Amplitude sweep flow curves of predicted formulation displayed a rheological behaviour consistent with the pilot dataset (Fig. 5). SPAAC hydrogels formulations still underwent crossover in the tested shear strain region, whereas DA hydrogels formulations did not.

With this reverse fabrication approach, it is possible to determine a priori the viscoelastic properties of hydrogels and bypass the laborious trial and error procedure of traditional research. This could have a crucial impact in designing 3D bioprinting protocols, especially for extrusion-based technologies. From hydrogel rheological properties it is possible indeed not only to optimize the printing process parameters (e.g. extrusion pressure, feed rate) but also to address material's behaviour after printing (e.g. shape fidelity, mechanical properties) [46].

The yield point (i.e. the minimum stress required for deformation) and the crossover point, which are intrinsically linked to both G′ and G″, reflect the number of cross-links within the hydrogel. Since elevated yield stress enhances filament formation and stiffness of the final construct, one should predict formulations with higher G′ values to develop hydrogels designed to be used as stiff ECM mimetics.

Table 7
Selected and experimental values G′ and G″ of SPAAC hydrogels predicted formulations.

	Selected values		Predicted formulations			Experimental values	
	G′ [Pa]	G″ [Pa]	Hyaluronic acid [mg]	Gelatin [mg]	4arm-PEG-DBCO [mg]	G′ [Pa]	G″ [Pa]
SP1	250.0	1.0	2.4	18.0	7.6	133.7	0.9
SP2	69.0	0.7	1.6	28.6	4.8	48.9	0.7
SP3	39.0	0.6	1.4	22.7	5.0	31.6	0.6
SP4	18.0	0.5	1.6	19.6	3.2	15.6	0.8

Table 8
Selected and experimental values G′ and G″ of DA hydrogels predicted formulations.

	Selected values		Predicted formulations			Experimental values	
	G′ [Pa]	G″ [Pa]	Hyaluronic acid [mg]	Gelatin [mg]	4arm-PEG-MA [mg]	G′ [Pa]	G″ [Pa]
DP1	20.0	2.0	2.8	27.0	13.7	26.2	2.9
DP2	12.0	1.5	2.0	27.6	10.8	10.3	1.4
DP3	5.0	1.0	1.6	25.1	8.2	3.7	0.9

Nonetheless, hydrogels with a yield stress too high will require extrusion pressure that might irreparably tear the material apart or build up a shear stress incompatible with cell viability. Following the same principle, to fabricate tender tissues, the rationale would be to predict hydrogels formulations with lower G′ values. In this case, even though extrusion from smaller nozzles might still be easily feasible, attention should be paid to avoid generating a runny unprintable liquid (G″ >> G′) [47,48].

3.4. 3D printability evaluation of predicted hydrogel formulations

The AI-predicted formulations were evaluated in terms of extrudability and printing accuracy, emphasizing their correlation with rheological parameters (i.e. G′ and G″).

The evaluation of ink extrudability from the nozzle was performed through a drop test, which consisted of increasing printing pressure until the formation of a droplet or a linear filament. As expected, hydrogels with higher storage modulus also correlated with a requirement for higher extrusion pressures (40–60 kPa). However, from Fig. 6A, B, C, D, and E it can be seen that only DP1 formulation managed to generate a continuous filament, whereas DP2, SP1, SP2 and SP3 formed much shorter filaments with a wide bulge at the edge. In contrast, DP3 and SP4 resulted in runny gels that just flowed straight out of the nozzle.

To evaluate shape fidelity of the 3D printed construct, an STL file was designed to obtain a one layer square with a zig-zag infill pattern (nominal strand thickness d = 1 mm) to facilitate the assessment of filament width, overlapping and gaps (Fig. 6H). Fig. 6I, L, and M presents the printing outcomes of DP1, DP2 and DP3, while Fig. 6N, O, P and Q depict SP1, SP2, SP3 and SP4 respectively. Despite all formulations being extrudable through the conical nozzle (G22), many differences can be pointed out. DP2 resulted to be the best formulation among DA hydrogels since it showed accurate strand thickness (d = 1,4 ± 0,1 mm, Sp % = 43 ± 15) (Fig. 6F and G) without gaps in the 3D construct. DP1 had the highest shape fidelity (d = 1,1 ± 0,3 mm, Sp % = 13 ± 26) but strands were irregular and curvy; moreover, evident gaps can be spotted. On the contrary, DP3 was not able to retain a 3D structure after printing.

For SPAAC hydrogels SP2 exhibited the best printing behaviour, producing a construct with still acceptable shape fidelity (d = 2,1 ± 0,4 mm, Sp % = 110 ± 40) and continuous filaments. Instead, SP1 revealed over-gelled ink features (Sp % = 21 ± 32), whereas SP3 and SP4 showed under gelation behaviour since no 3D shape could be retained after printing [49].

Therefore, from these printing tests it seems clear that formulations with higher storage modulus, DP1-DP2 for DA hydrogels and SP1-SP2 for SPAAC hydrogels respectively, exhibited better printing accuracy, as their elastic properties allow preservation of structural integrity during and after extrusion. Conversely, lower G′ values led to poorly defined shapes as observed in SP3, SP4, and DP3 3D constructs. This trend also correlated with ink swelling behaviour (Supporting information Fig. S5).

The fact that DA hydrogels require a lower absolute G′ compared to SPAAC hydrogels to obtain an accurate 3D construct, can be explained

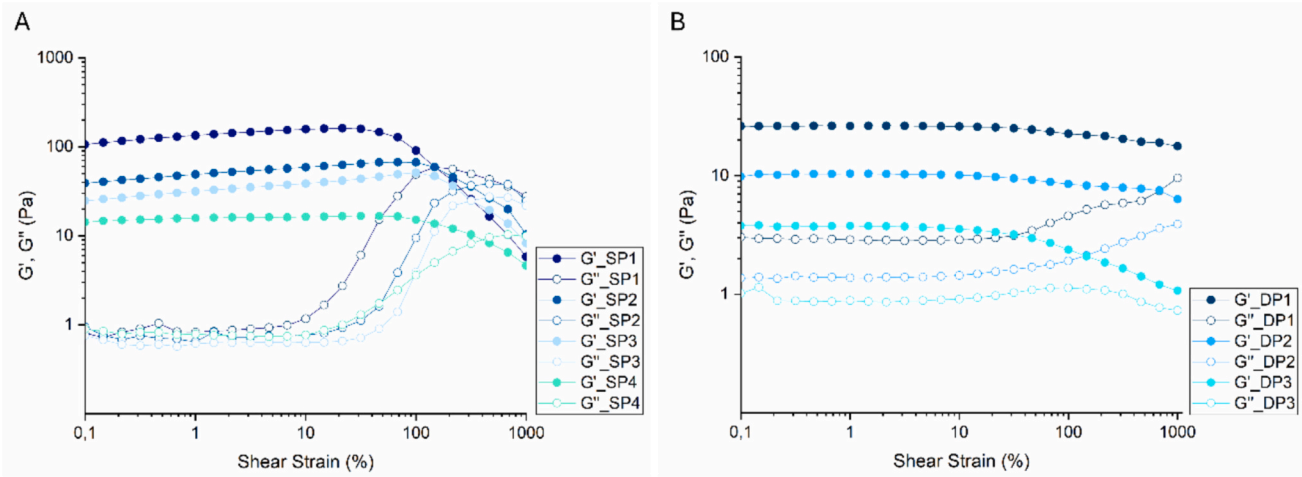

Fig. 5. Amplitude sweeps of predicted hydrogel formulation of SPAAC (A) and DA (B).

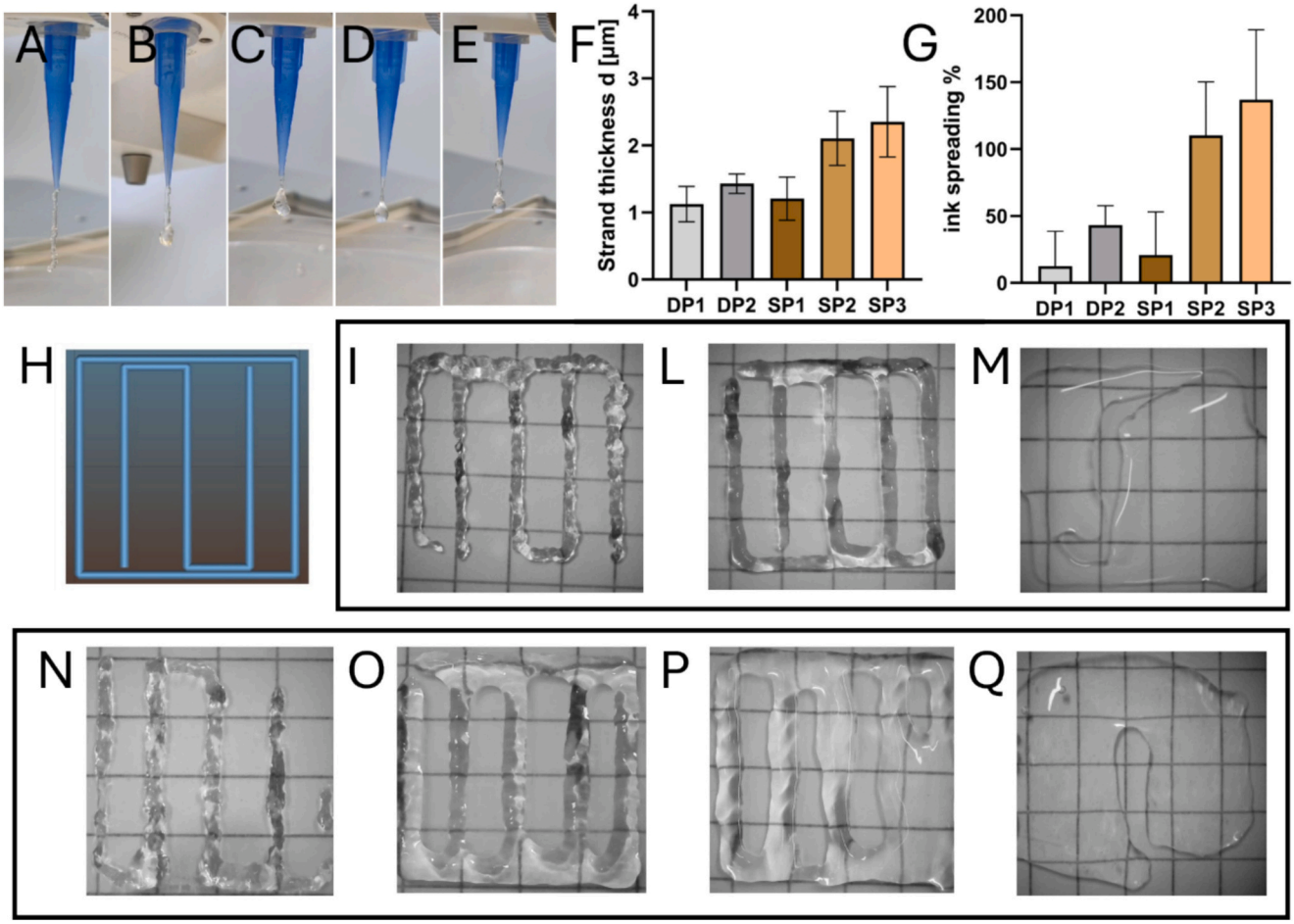

Fig. 6. Drop test of DP1 (A), DP2 (B), SP1 (C), SP2 (D), and SP3 (E); (F) strand thickness; (G) and ink spreading factor of 3D-printed formulations, (H) CAD used for 3D printing; Images of 3D construct made with DP1 (I), DP2 (L), DP3 (M), SP1 (N), SP2 (O), SP3 (P), and SP4 (Q).

by two factors: linker length and G″ values. As discussed above, DA hydrogels have a more compact structure that favours intramolecular forces. This in turn, confers the ink a solid-like behaviour (G′ > G″) across a wider shear strain range [41,42]. Moreover, DP1 and DP2 hydrogels have loss tangent (tan δ = G″/G′) values (tan δ ~0,1) greater than SP1 and SP2 hydrogels (tan δ ~0,01). In a recent work, Gao and colleagues, indeed proposed a model to assess ink printability, where higher loss tangent values were correlated with improved extrusion uniformity [50].

4. Conclusions and future directions

A user-friendly predictive tool based on ML designed to facilitate and expand the generation of extracellular matrix (ECM) mimics with

tailored rheological and biological properties has been proposed and validated. The ML models that we developed were capable of accurately predicting hydrogel formulations starting from their rheological properties (i.e. G′ and G″), enabling the customization of biomaterials starting from desired rheological properties without the extensive trial-and-error experimental procedure traditionally employed in the development of ink for 3D bioprinting applications.

In our opinion, this predictive platform represents great innovation with respect to other ML models present in the literature. Moreover, key highlights include the use of natural polymers that bear bioactive properties (e.g. cell adhesion, cell fate modulation), macromolecules of different nature with different reactivity (i.e. a protein and a polysaccharide), as well as crosslink via click chemistry reactions which allow for an accurate biomolecular and physical control with direct effect on cell fate regulation. Another important aspect is the use of a limited training dataset by relying on both the generation of a synthetic dataset and practical constraints within which hydrogel formation is possible and/or useful.

These advancements offer new solutions to speed up the research on the development of next-generation biomaterials for complex biomedical applications, expanding the use of bioactive hydrogels for multidisciplinary laboratories and their application in clinical translation. Future direction in AI-assisted research for tissue mimetic models should focus on standardizing data collection and most importantly on sharing knowledge among research groups in order to enhance ML models training, and therefore predictive accuracy. In this way, it would be also possible to build a unified model capable of predicting multiple materials formulations, composed of several macromolecules and crosslinked with different chemoselective reactions. The integration of rational design principles and data selection with state-of-the-art instrumentation in the laboratory routine will have a tremendous impact on academic research that focuses on biomedical fields.

CRediT authorship contribution statement

Francesca Cadamuro: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation. **Marco Piazzoni:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation. **Elia Gamba:** Validation, Methodology, Investigation, Formal analysis, Data curation. **Beatrice Sonzogni:** Writing – review & editing, Investigation. **Fabio Previdi:** Writing – review & editing, Supervision, Funding acquisition. **Francesco Nicotra:** Writing – review & editing, Funding acquisition. **Antonio Ferramosca:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Laura Russo:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bioadv.2025.214323.

Data availability

Data will be made available on request.

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